

A Conceptual Model for Human-AI Collaboration in Managerial Work

Abstract

Managerial work involves strategic, timely, and complex decision making, impacting employees, strategy, and the economy of the company. While AI systems can assist by integrating complex data and facilitating timely decisions, managers may face challenges with transparency and risk losing agency over their decisions. This paper presents a two-dimensional conceptual model mapping AI systems along the augmentation-automation dimension (agency distribution) and analysis-synthesis dimension (cognitive orientation). Using a design science approach based on literature, we identify four distinct groups, each presenting unique implications for AI systems and managerial roles and competencies. The framework advances the theory of human-AI collaboration while providing practitioners with actionable guidance for system selection and implementation. By revealing strategic choices in AI design, organizations can align technology adoption with managerial responsibilities, preserving human agency while enhancing organizational capabilities.

Keywords: AI systems, Human-AI collaboration, Managerial work, System design, Organizational implementation

1. Introduction

The pervasive influence of artificial intelligence (AI) holds significant potential for reshaping organizational landscapes, as businesses increasingly explore AI applications to enhance or even redefine traditional managerial tasks [1]. Despite the growing interest, many AI applications remain confined to experimental pilots and often struggle to integrate effectively into broader managerial workflows. This implementation challenge reflects a fundamental mismatch between AI system capabilities and the distinctive requirements of managerial work. Unlike operational tasks with

clearly defined inputs and outputs, managerial activities involve complex decision making under uncertainty, coordination across organizational boundaries, and navigation of competing stakeholder interests. These characteristics create unique challenges for AI system design that existing frameworks fail to address.

Recent studies show how AI's evolving role in management blurs the line between tool and teammate, fostering a form of 'conjoined agency' where humans and AI share responsibilities [2]. Despite this evolving relationship, much of the literature treats AI as a uniform 'black box,' obscuring crucial distinctions and preventing the development of targeted design guidance [3, 4]. This lack of specificity leaves practitioners without systematic methods to assess which AI systems align with specific managerial tasks or anticipate the organizational implications of different technological choices.

The theoretical literature on human-AI collaboration predominantly offers broad principles rather than actionable frameworks that address the specific design challenges of managerial AI systems. Existing taxonomies focus on collaboration styles or business value but fail to connect these dimensions to the cognitive nature of managerial work itself, whether it is reductive (analysis) or generative (synthesis) [5]. This gap between theory and practice limits both our understanding of how AI transforms managerial work and practitioners' ability to make informed system selection retrospectively, or to guide future system design decisions.

This paper addresses these limitations by developing a conceptual model that maps AI systems along two critical dimensions: the augmentation-automation dimension, which captures how systems distribute agency between managers and machines, and the analysis-synthesis dimension, which distinguishes between refining and creative cognitive functions. This two-dimensional framework creates four distinct quadrants, each representing different approaches to human-AI collaboration with unique implications for

transparency, control, and skill requirements.

2. Related Works

The development of AI systems for managerial contexts builds on research on human-machine interaction in decision-making. This section reviews key frameworks for human-AI collaboration and taxonomies of cognitive tasks and AI competencies that support our conceptual model. We focus on literature addressing the distribution of agency between humans and machines, task classification, and the skills needed for effective AI-human collaboration in organizations.

2.1. Taxonomies of Human-AI Collaboration and AI Applications

Schröder et al. (2022) offer a taxonomy of human-AI collaboration patterns (supervised assistance, shared control, autonomous delegation), each with differing transparency and trust requirements. Their classification clarifies interaction modes but stops short of linking those modes to underlying AI architectures or specific managerial competences [6]. Engel et al. (2022) develop a taxonomy of AI use cases based on the organizational business value they deliver, distinguishing strategic, operational, analytical, and customer-facing applications [7]. They argue that aligning AI initiatives with clear value dimensions helps organizations calibrate the balance between automation and human oversight. This taxonomy highlights how various AI configurations can either support or supplant managerial tasks in different business domains. Jarrahi and Glaser propose a detailed typology of AI systems, highlighting the distinct human activities required for each category's implementation. Their framework underscores how differences in learning paradigms and material properties influence both organizational impacts and the specific work activities needed for effective integration [8].

While these frameworks offer crucial insights into collaboration patterns [6], dynamic initiative [9, 10], and value alignment [7], they do not explicitly connect these dimensions to the cognitive nature of the managerial task itself, whether it is reductive (analysis) or generative (synthesis). Jarrahi and Glaser's typology links AI to human activities but does not map these onto a continuous space of agency from augmentation to automation. Our model addresses this gap by integrating both the distribution of agency and the cognitive orientation of the task, providing a more comprehensive framework for system design and managerial role analysis.

2.2. Foundations of Managerial Tasks and Design Tensions in AI Systems

Integrating AI into managerial work presents unique challenges due to the inherent ambiguity, interdependence, and contextual complexity that distinguish it from more structured operational tasks [11]. Recent research adopts a task-centered view that breaks managerial work into discrete activities with identifiable inputs, outputs, and opportunities for AI integration [3, 12]. From these, we identify key managerial activities illustrating AI potential: performance monitoring and reporting, decision analysis and forecasting, strategic planning and resource allocation, team coordination and communication, talent and knowledge management, and innovation and problem solving. This task-based framework precisely maps managerial functions to AI capabilities and reveals that identical outcomes can emerge from different types of AI implementations. The design of AI systems for managerial tasks involves balancing competing values and capabilities. Automating analytical tasks surfaces the efficiency versus engagement tension, where streamlined processes can improve productivity, but may also reduce managerial oversight and risk complacency [1]. Tools that synthesize novel outputs introduce the creativity versus governance tension, requiring a balance between algorithmic generation and organizational constraints [13]. These tensions highlight that integrating AI is not just about adding functionality but about navigating the balance between automation and human judgment, which depends on the nature of managerial tasks.

This perspective offers two key advantages. First, it connects managerial tasks with their automation potential and helps clarify where AI adds value. Second, it defines design targets by distinguishing whether AI should provide analytical support, synthesis assistance, process augmentation, or full automation. Different task-AI configurations require distinct managerial skills: analytical AI demands data interpretation abilities, while synthetic AI requires formulating effective prompts and critically evaluating AI-generated outputs. This highlights that effective AI integration increasingly demands AI literacy and technical skills that fall outside traditional managerial functions [14]. Recognizing these task characteristics is crucial as they determine the specific AI literacy, cognitive focus, and power delegation required for effective manager-machine collaboration.

2.3. The Augmentation-Automation Dimension

A core design tension in AI systems for managerial use lies in how responsibility and agency are distributed between human and machine. This tension is commonly described as the augmentation-automation paradox, which is the idea that AI can either extend human discernment or substitute for it, but rarely achieves both without trade-offs [15]. Excessive automation in domains requiring discretion can undermine human engagement, trust, and accountability, whereas thoughtful augmentation can enhance performance without displacing agency [16, 17]. Augmentation refers to systems that enhance human decision-making by providing insight, options, or guidance while leaving control with the manager. Automation, in contrast, shifts execution or decision authority to the system itself, often reducing the manager's direct involvement [10]. Importantly, this continuum is not binary but a spectrum: real-world AI systems rarely occupy purely one end or the other. Instead, they often automate routine processing while augmenting strategic tasks, or shift between modes as complexity changes.

This continuum represents a context-sensitive control mechanism that designers can tune through override controls, transparency indicators, and feedback loops [18]. Dellermann et al. (2019) argue for adaptive autonomy in hybrid intelligence systems, where automation levels dynamically adjust based on changing data, workload, or user preferences [5]. This adaptive perspective transforms the augmentation-automation dimension from a static classification into a flexible design lever that responds to contextual demands rather than imposing fixed boundaries between human and machine responsibilities [19].

Building on this adaptive perspective, Muller and Weisz (2022) proposed a model that differentiates levels of automation along a spectrum from full human control to full machine autonomy. Central to their framework is the distinction between humans being in-the-loop, where managers directly monitor and guide decisions in real-time, and on-the-loop, where their role shifts to supervisory oversight with intervention triggered only when exceptions occur [9]. This distinction illuminates how managerial engagement can dynamically shift between hands-on involvement and strategic supervision, ensuring AI systems remain aligned with organizational goals while adapting to changing operational conditions [20]. This dynamic redistribution of agency fundamentally transforms managerial work itself, as managers increasingly allocate cognitive resources toward interpreting AI

outputs and maintaining algorithmic oversight rather than engaging in direct interpersonal coordination with their teams. This shift redefines both the competencies required and the relational foundation of management practice.

2.4. The Analysis-Synthesis Dimension

The Analysis-Synthesis axis captures two complementary modes of cognitive work that define how managers engage with information and generate solutions. Analysis-oriented tasks involve deconstructing complex problems into constituent elements, identifying patterns within existing data, and extracting meaning from information structures [21]. Synthesis-oriented tasks, conversely, involve recombining disparate information sources, generating novel solutions, and integrating fragmented elements into coherent strategic outputs [22]. This distinction aligns with established cognitive frameworks: analysis corresponds to convergent thinking processes (categorization, evaluation, problem decomposition), while synthesis aligns with divergent thinking (ideation, creative recombination, strategic integration) [23].

Recent work on human-AI collaboration has begun addressing this cognitive dichotomy. Song et al. (2024) introduce a "cognitive mode" dimension distinguishing analysis-oriented from synthesis-oriented AI agents [22]. They define analysis-oriented AI as performing reductive data processing such as pattern recognition, classification, and predictive modeling, which transforms complex information into more manageable and meaningful forms. Synthesis-oriented AI, on the other hand, focuses on generative tasks like concept development, design optimization, and creative ideation that integrate and enrich existing information.

These cognitive orientations impose distinct design requirements on AI systems. Analytic tasks demand systems emphasizing transparency, traceability, and interpretability. Managers using AI for analysis require clear explanations, detailed visualizations, and drill-down capabilities to validate system inferences and maintain confidence in data-driven conclusions. Synthetic tasks, however, prioritize generative capacity, option diversity, and iterative refinement. Managers engaging with synthesis-oriented AI need systems that produce multiple alternatives, support modification and customization, and enable creative exploration without prescriptive constraints.

3. Data Collection and Conceptual Development

This research employs a design science approach to develop a conceptual framework to systematically categorize and analyze AI systems designed for managerial contexts. Design science research focuses on creating artifacts that solve identified organizational problems while contributing to both practical and theoretical knowledge [24]. For constructing the framework, we employed a two-phase approach combining targeted literature review with retrospective case validation [25]. The framework development process involved identifying key tensions in AI system design through iterative analysis of existing literature, then validating the resulting model through systematic mapping of deployed systems. We identified research streams relevant to AI systems and managerial work, focusing on human-AI interaction, AI literacy or managerial competence, and AI system design. Rather than attempting exhaustive coverage, we used a purposive sampling strategy prioritizing theoretical depth and significance of perspectives [26]. To validate the model's explanatory potential and practical applicability, we conducted retrospective mapping of four representative AI systems currently deployed and applicable in managerial contexts: Microsoft Viva Insights, Tableau's Ask DataLens, Oracle Financial Analytics, and Workday Adaptive Planning. These systems were selected to represent diverse technical architectures, and implementation contexts covering all quadrants of the model. For each system, we analyzed publicly available product documentation, implementation cases, and user interface specifications to determine positioning along both dimensions, following established design science research methods for artifact demonstration [27, 24].

During the literature synthesis phase, we applied a constant-comparison coding scheme to identify latent tensions shaping the design of AI systems in managerial contexts [28]. We examined each source for explicit or implicit distinctions regarding how AI systems distribute control between humans and machines and how they differ in their cognitive orientation toward analytical versus generative tasks. Through iterative coding and refinement, two theoretically robust dimensions emerged: the augmentation-automation dimension representing agency distribution and the analysis-synthesis distinction capturing cognitive orientation.

Because managerial AI systems face distinct requirements around transparency, accountability, and organizational integration that differ from

operations-focused AI tools or consumer applications, we established specific criteria for system selection. Selected systems had to be enterprise software solutions that support managerial use cases and that are adequately documented to allow us to assess their functionality and position within the framework.

We deliberately selected one representative system per quadrant to illuminate clear conceptual distinctions. Each chosen system demonstrated a strong, unambiguous fit within its quadrant, helping us to show how the framework can distinguish between different types of AI systems and clarify their unique capacities in practice.

4. Conceptual Framework for Managerial AI Integration

This conceptual framework presents a two-dimensional model that maps AI systems for managerial work along agency distribution (augmentation-automation) and cognitive orientation (analysis-synthesis) dimensions, revealing strategic design choices and their organizational implications. To apply the framework effectively, practitioners begin by identifying a specific managerial task such as performance monitoring, strategic planning, or resource allocation. This task-first methodology ensures that AI solutions align with actual work requirements rather than driving adoption based solely on technical specifications. Once a target task is selected, practitioners systematically identify existing AI systems designed to address that specific managerial task. Each identified system is then mapped onto the two-dimensional model on its design characteristics. This mapping process reveals the strategic choices inherent in different technological configurations to the same fundamental managerial task, thus enabling informed decision-making about which AI approach best aligns with organizational values, managerial capabilities, and strategic objectives.

The intersection of these dimensions creates a structured design space that enables precise alignment of AI capabilities with specific managerial cognitive demands. Beyond retrospective mapping, system architects can use the model prospectively when designing new managerial AI tools. Mapping a proposed system within a specific quadrant informs key architectural choices about transparency mechanisms, control allocation, and interface design. Depending on how AI systems designers adjust these dimensions, systems can prioritize either analytical insight extraction or synthetic solution generation while maintaining appropriate levels of

human involvement. This capacity to support both retrospective analysis and forward-looking design highlights the framework's methodological value for advancing AI system development and research on organizational implementation.

5. Applying the Conceptual Model to Managerial Tasks

Mapping AI systems across our two-dimensional model reveals how different configurations fundamentally reshape the execution of specific managerial tasks. Each quadrant represents a distinct approach to distributing cognitive labor and decision authority between humans and machines, creating unique implications for managerial roles, organizational integration, and system design requirements. To illustrate this principle, we examine performance monitoring, a core managerial responsibility involving the tracking, analysis and response to organizational and team performance metrics.

5.1. Example: Performance Monitoring Across Four AI Configurations

By examining how different AI systems approach performance monitoring, we reveal the strategic choices inherent in AI system design and their consequences for managerial work. Each quadrant represents a distinct philosophy of human-machine collaboration applied to the same core task, demonstrating how technological positioning fundamentally shapes both system capabilities and user requirements.

The Augmentation-Analysis quadrant features AI systems designed to support managers by deconstructing complex data into actionable insights, while final judgment and decision authority remain firmly in human hands. This approach directly enhances core managerial tasks like performance monitoring and decision analysis. Instead of delivering prescriptive conclusions, these systems provide analytical support while managers act as pattern analysts that surface non-obvious patterns and anomalies, typically through intelligent dashboards and exploratory interfaces [29]. Microsoft Viva Insights exemplifies the Augmentation-Analysis approach to performance monitoring. It enhances managers' analytical capabilities by performing intensive computational processing, such as deconstructing communication data to surface collaboration patterns and potential workload imbalances. This system delivers insights and identifies non-obvious patterns, serving as an analytical partner. The central principle in this quadrant is a human-AI symbiosis: while the system

handles complex data analysis, the manager provides the essential contextual knowledge and qualitative judgment to interpret these outputs within their team's unique dynamics and to validate findings before taking action, thereby preserving human agency [16]. This AI system preserves traditional managerial agency while enhancing analytical capabilities, requiring managers to develop skills in data interpretation and contextual analysis.

In the Augmentation-Synthesis quadrant, AI acts as a collaborative partner, generating diverse strategic alternatives and creative solutions, such as scenario plans, resource allocations, or process redesigns, for managers to evaluate, refine, and integrate based on organizational context and priorities [30]. These systems offer multiple options or drafts, not single solutions, thereby encouraging managerial input and embodying collaborative intelligence where human and machine strengths enhance the ideation process. Applied to performance monitoring, Tableau's Ask DataLens feature exemplifies this approach by enabling managers to generate multiple analytical perspectives and performance narratives from the same underlying data. The system allows managers to pose natural language questions about performance trends, which then automatically generates diverse visualizations, comparative analyses, and exploratory views. This system configuration transforms managers into collaborative creators with AI in generating performance insights, demanding competencies in query formulation and synthetic evaluation.

The Automation-Analysis quadrant comprises AI systems that autonomously execute full analytic pipelines, yielding ready-made results for managers to monitor. These systems take over routine analytical work, such as autonomous performance monitoring and exception reporting platforms, freeing managers to focus on exception handling and higher-level concerns [31]. In this quadrant, tools like Oracle Financial Analytics automatically assess budgetary performance, flag deviations, and generate standardized reports without manager input. The system continuously monitors database performance metrics, automatically detects anomalies using machine learning algorithms, and provides standardized reports highlighting issues that require attention, with managers receiving alerts only when performance thresholds are exceeded or when the system identifies patterns that warrant human review. This system configuration is characterized by high automation in routine analytical tasks and typically include override mechanisms for exceptional cases. Essentially repositioning managers as exception handlers and strategic overseers, requiring skills in

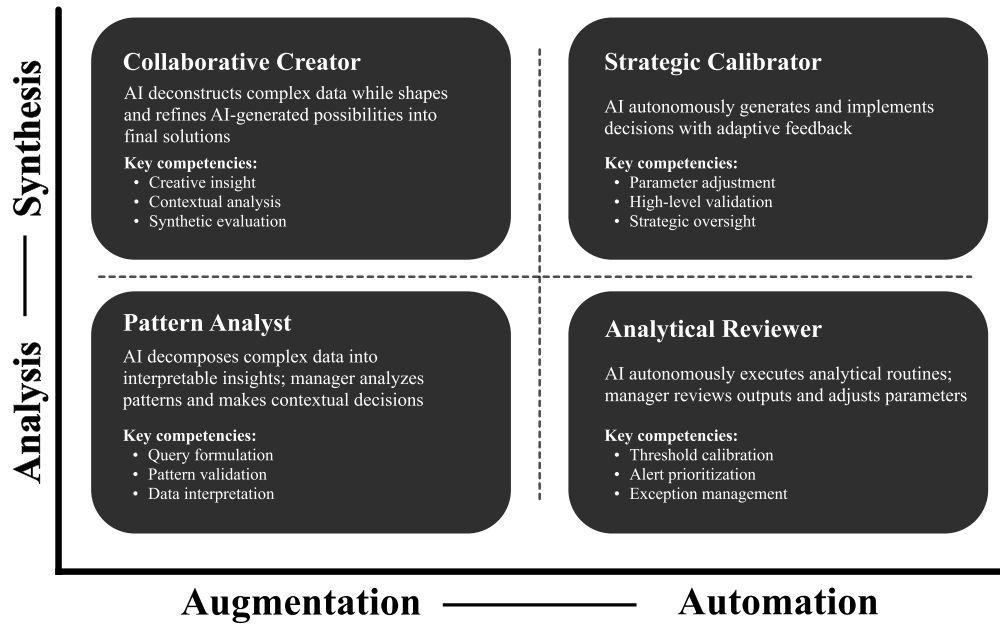


Figure 1. Conceptual framework mapping AI systems along augmentation-automation (horizontal) and analysis-synthesis (vertical) dimensions, defining four quadrants with distinct managerial roles, system characteristics, and skills.

threshold calibration and alert prioritization.

The Automation-Synthesis quadrant represents the most autonomous class of AI, featuring systems that independently generate and implement novel decisions or artifacts through closed-loop operations. These self-adjusting optimization systems synthesize and execute solutions without requiring direct managerial intervention. Workday Adaptive Planning's continuous planning capabilities exemplify this approach to performance monitoring by automatically generating updated performance forecasts, budget adjustments, and resource allocation recommendations based on real-time performance data. This design repositions managers as above-the-loop actors, essentially as conductors who oversee AI systems at the operational level. Rather than engaging in collaborative synthesis, managers in this quadrant focus on strategic calibration, managing exceptions, and evaluating autonomous outcomes after implementation [9]. Managerial tasks shift from hands-on co-creation to high-level supervision, focusing on evaluating AI-generated decisions post-implementation and intervening when outcomes diverge from organizational objectives. This system configuration elevates managers to strategic calibrators of autonomous systems, demanding competencies in parameter adjustment and high-level validation.

System designs uniquely affect managerial tasks, trust, and organizational outcomes. Each configuration

creates distinct organizational implications. Augmentation-analysis systems maintain existing accountability structures while potentially increasing cognitive load. Automation-synthesis systems enable creative exploration of performance data while requiring investment in query formulation skills. These systems improve efficiency in routine monitoring while risking managerial disengagement from performance details. Automation-analysis systems maximize operational autonomy while potentially creating distance between managers and the performance monitoring process itself. Understanding these implications enables organizations to select AI configurations that align with their values, capabilities, and strategic objectives rather than adopting AI systems based solely on technical capabilities or cost considerations. The framework reveals that the choice of AI configuration for performance monitoring is ultimately a choice about how to distribute agency, cognitive responsibility, and accountability within the organization.

6. Discussion

This paper presents a conceptual framework that reveals the strategic diversity inherent in AI systems designed for managerial contexts. The two-dimensional model illuminates how different system architectures fundamentally reshape the distribution of cognitive responsibility and decision authority between humans

Quadrant & System Archetype	Description	Key System Characteristics	Core Design Tensions	Managerial Competencies Required	Opportunities	Challenges
Augmentation - Analysis Pattern Analyst	AI supports managers by deconstructing data; human retains oversight	Delivers unbiased insights without compromising authority	Transparency vs. Complexity, Objectivity vs. Intuition	Analytical reasoning and evaluation of AI outputs	Improved decision quality, autonomy preserved, pattern discovery	Cognitive overload, data misinterpretation, analysis paralysis
Augmentation - Synthesis Collaborative Creator	AI collaborates to generate novel content; human curates, selects outcomes	Creative partner offering multiple options; manager retains editorial control	Creativity vs. governance, Inspirations vs. Ownership	Creative evaluation, refinement, and curation of AI outputs	Amplified creativity, broader solution space	Risk of narrow AI suggestions, requires human refinement, output dependency
Automation - Analysis Analytical Reviewer	AI performs and executes analytic routines; manager monitors results	Automated pipelines with exceptions delegated to humans	Efficiency vs. Engagement, Volume vs. Meaningfulness	Alert management and oversight of automated outputs	Streamlined tasks, strategic focus enabled	Manager complacency, reduced vigilance, alert fatigue
Automation - Synthesis Strategic Calibrator	AI autonomously generates and implements decisions/artifacts	End-to-end automation with adaptive feedback loops	Autonomy vs. Control, Innovation vs. Risk	Meta-management, fine-tuning AI decision processes	Scalable operations, high efficiency, self-adaptation	Black-box risks, managerial deskilling, drift detection

Table 1. Conceptual model categorizing AI systems along augmentation-automation and analysis-synthesis dimensions. Quadrants define characteristics, tensions, competencies, opportunities, challenges.

and machines. By mapping AI systems along the augmentation-automation and analysis-synthesis dimensions, the model provides a structured lens for understanding how technological choices create distinct pathways for human-AI collaboration in managerial work. The framework’s practical value lies in its ability to make visible the often implicit design decisions that determine how AI systems integrate with managerial tasks and organizational values. Each quadrant represents not merely a technical configuration, but a distinct approach to redistributing agency and cognitive labor within management practice. This perspective opens important considerations about how AI adoption transforms managerial roles, reshapes organizational accountability structures, and demands new forms of competency development that align with specific system characteristics rather than generic technological capabilities.

6.1. Opportunities and Constraints Across the Conceptual Model for Managers and Organizations

Each quadrant presents distinct opportunities and challenges for managers. Augmentation-Analysis configurations enhance decision quality by delivering nuanced insights while preserving managerial autonomy and accountability [15]. Yet richer information flows may also increase cognitive burden. Designers must therefore ensure that interfaces distill insights into actionable outputs without overwhelming users [32, 18]. By contrast, Automation-analysis excel at streamlining routine tasks, allowing managers to redirect focus

toward strategic initiatives [6]. However, this quadrant may introduce what Bainbridge (1983) called an irony of automation: the more effective the AI becomes at autonomously handling routine analysis, the greater the risk of managerial complacency and erosion of vigilance [33]. This can leave managers ill-prepared to handle the very exceptions the system escalates to them. To mitigate this tension, systems must be designed not just for automation but for sustained engagement, incorporating features like transparent alert mechanisms, contextual escalation pathways, and “fire drills” that require managers to periodically engage with the underlying data, ensuring they retain critical oversight capabilities [34, 10].

Augmentation-synthesis tools amplify creative ideation by helping managers move beyond fixed mindsets and explore broader solution spaces [35]. However, when AI suggestions lack diversity, they risk reinforcing narrow perspectives. To preserve creative breadth, designers should offer options to customize or refine AI-generated proposals in ways that remain accessible to non-technical users [36, 37]. Lastly, Automation-synthesis systems extend managerial reach by automating both creative and operational tasks [29]. Yet their opacity raises concerns around trust and expertise erosion. Transparency features, override controls, and feedback mechanisms are essential to ensure managers retain meaningful involvement and continue to develop their judgement [15, 18]. Taken together, these quadrant-specific insights help organizations weigh not just what AI systems do, but how they reshape the roles and responsibilities of those

who use them.

6.2. Rethinking Managerial Work and Task Distribution

Adopting an AI system involves more than a change in technology; it demands a fundamental reconsideration of managerial work and which tasks managers perform versus delegate to automated systems. Depending on where a system sits within the model, it reshapes the composition and execution of core managerial activities. For instance, in the augmentation-analysis quadrant, AI transforms performance monitoring tasks by surfacing data patterns that managers must still interpret and contextualize. In contrast, automation-synthesis deployments shift managers away from routine strategic planning activities toward exception handling and high-level calibration work.

Our model highlights how each AI implementation redistributes authority, cognitive responsibility, and accountability. This distinction is vital, as organizations often codify managerial duties in job descriptions, performance metrics, and cultural norms. If AI changes these responsibilities by reducing analytical work or replacing creative roles, it risks conflicting with organizational values such as human judgment, ethical oversight, and team cohesion.

Through this conceptual lens, any new AI adoption at the managerial layer should prompt examination of how core work activities will be redistributed. For instance, a company dedicated to bottom-up, collaborative decision-making may find that automating strategic planning tasks undermines their established workflow of participatory decision-making. Similarly, organizations that emphasize a culture of human ingenuity may resist automating core strategic tasks. By offering a shared vocabulary for positioning AI systems along two system dimensions, the model enables stakeholders to surface potential conflicts before deployment. Designers and executives can ask: Does this tool prioritize machine autonomy in ways that undermine distributed leadership? Or does it preserve enough transparency and human-in-the-loop controls to stay congruent with our governance principles?

Using the model effectively requires examining not only what tasks an AI tool performs, but how it reconfigures the essential work activities that define effective management in the organization. This kind of structured assessment can surface tensions that a simple ROI analysis might overlook. A system positioned in the automation-analysis quadrant may improve efficiency in budget monitoring tasks, but if

managers no longer engage in shared budget reviews with their teams, the organization risks losing shared accountability in financial oversight. Conversely, an augmentation-synthesis tool that generates marketing ideas may support a brand's innovation ethos, yet raise new concerns around intellectual property and content oversight. In both cases, the model helps stakeholders clarify how AI shifts responsibility and whether those shifts align with the organization's values and governance norms.

By providing a structured framework and shared terminology, the model supports concrete, actionable analysis of how AI adoption enhances rather than fragments the coherent execution of managerial work, preserving essential coordination and oversight activities while gaining efficiency in routine tasks. It helps organizations translate broad intentions into specific questions, such as, "Which quadrant does this system occupy, and how will that affect the manager's role in relation to our value of shared accountability?" It enables organizations to select AI tools that enhance performance while aligning with their cultural and ethical priorities, both now and into the future. Ultimately, rethinking managerial tasks through these dual dimensions ensures that AI adoption is not just a technical upgrade, but a values-driven transformation that reinforces, rather than erodes, the organization's foundational code.

6.3. AI Literacy and Managerial Competences

AI literacy for managers cannot be treated as a uniform skill set; it must be understood as a dynamic, context-sensitive competence shaped by the specific AI system and, more importantly, the managerial tasks it transforms [14]. Rather than applying generic competencies uniformly across technologies [38], our framework reveals that managers need targeted skills aligned with their system's quadrant.

Each quadrant demands distinct competencies. Augmentation-analysis systems require managers to interpret analytical outputs, validate insights, and contextualize findings within organizational dynamics. Augmentation-synthesis systems necessitate skills in evaluating generative content, refining AI-produced alternatives, and maintaining creative oversight. Automation-analysis systems demand competencies in exception monitoring, threshold calibration, and maintaining analytical vigilance despite reduced hands-on involvement. Automation-synthesis systems require skills in strategic parameter adjustment, outcome validation, and high-level system calibration [39].

This quadrant-specific approach reflects the technical diversity underlying AI systems. Rule-based systems require managers to grasp predefined logic and decision trees, while generative AI demands prompt engineering and critical content evaluation. Supervised learning models necessitate competencies in data validation, while reinforcement learning systems require understanding of reward structures and feedback mechanisms [40].

By mapping specific managerial tasks such as performance monitoring, decision analysis, strategic planning, and team coordination to designated quadrants, the framework demonstrates that AI literacy skills must align with the cognitive demands and execution patterns characterizing each category of managerial activity. This targeted approach ensures managers develop capabilities that match their specific AI system interactions rather than pursuing generic technical knowledge.

7. Concluding Remarks

This paper presents a conceptual model that maps AI systems in managerial work, spanning the augmentation-automation and analysis-synthesis dimensions. We adopt a sociotechnical perspective that demonstrates how AI value depends not only on technical capabilities but also on how systems integrate with specific managerial task requirements. For AI system practitioners, the model guides alignment of system functionality with distinct managerial activities, while determining appropriate levels of task automation versus human oversight. For researchers, it establishes a foundation for exploring how different AI configurations reshape the execution of core managerial tasks and transform the competencies required for effective task performance. By moving beyond monolithic views of AI, the model enables deliberate, context-sensitive development of tools that enhance rather than displace essential managerial work activities.

8. Limitations

While offering a structured lens, our conceptual model has three key limitations. First, the framework relies on retrospective analysis of existing systems rather than empirical validation across diverse organizational contexts, limiting generalizability claims. Second, the model centers on the manager-AI interaction and does not address broader organizational dynamics such as team coordination, stakeholder alignment, or cultural adaptation. Third, many AI systems combine features across quadrants, making strict categorization difficult

and requiring more flexible, feature-level application in practice.

9. Future Research Directions

This framework opens several promising directions for future research. First, empirical studies across varied organizational contexts are needed to test the model's generalizability and refine its archetypes. Cluster analyses of deployed AI tools may reveal hybrid forms of human-AI collaboration and clarify links to business outcomes. Second, longitudinal research should examine how systems and managerial tasks co-evolve, shedding light on factors that drive quadrant shifts and their implications for managerial agency. And third, design-oriented work should address core tensions within each quadrant, such as maintaining human oversight in automated systems or supporting creativity in generative tools.

References

- [1] C. Dietzmann and Y. Duan, "Artificial Intelligence for Managerial Information Processing and Decision-Making in the Era of Information Overload," 2022.
- [2] A. Murray, J. Rhymer, and D. G. Sirmon, "Humans and Technology: Forms of Conjoined Agency in Organizations," *Academy of Management Review*, vol. 46, pp. 552–571, July 2021.
- [3] K. C. Kellogg, M. A. Valentine, and A. Christin, "Algorithms at Work: The New Contested Terrain of Control," *Academy of Management Annals*, vol. 14, pp. 366–410, Jan. 2020.
- [4] C. Curchod, G. Patriotta, L. Cohen, and N. Neysen, "Working for an Algorithm: Power Asymmetries and Agency in Online Work Settings," *Administrative Science Quarterly*, vol. 65, pp. 644–676, Sept. 2020.
- [5] D. Dellermann, A. Calma, N. Lipusch, T. Weber, S. Weigel, and P. Ebel, "The Future of Human-AI Collaboration: A Taxonomy of Design Knowledge for Hybrid Intelligence Systems," 2019.
- [6] A. Schroder, I. Constantiou, V. Tuunainen, and R. D. Austin, "Human-AI Collaboration – Coordinating Automation and Augmentation Tasks in a Digital Service Company," 2022.
- [7] C. Engel, J. Schulze Buschhoff, and P. Ebel, "Structuring the Quest for Strategic Alignment of Artificial Intelligence (AI): A Taxonomy of the Organizational Business Value of AI Use Cases," 2022.
- [8] V. Glaser and M. H. Jarrahi, "Not All AI Systems Are Created Equal: A Typology of AI Systems' Performance and Affordance," 2025.
- [9] M. Muller and J. Weisz, "Extending a Human-AI Collaboration Framework with Dynamism and Sociality," CHIWORK '22, (New York, NY, USA), pp. 1–12, Association for Computing Machinery, 2022.
- [10] B. Shneiderman, "Human-Centered AI: A New Synthesis," in *Human-Computer Interaction – INTERACT 2021*, pp. 3–8, Springer International Publishing, 2021.

- [11] M. H. Jarrahi, G. Newlands, M. K. Lee, C. T. Wolf, E. Kinder, and W. Sutherland, "Algorithmic management in a work context," *Big Data & Society*, vol. 8, p. 20539517211020332, July 2021.
- [12] I. Lippert, K. Kirchner, and M. Wiener, "Context Matters: The Use of Algorithmic Management Mechanisms in Platform, Hybrid, and Traditional Work Contexts," 2023.
- [13] I. Sudeeptha, W. Müller, A. Richter, M. Leyer, and F. Nolte, "Obstacles to Human-AI Collaboration," 2024.
- [14] J. Locke and K. Rainer, "Teaching Artificial Intelligence and Machine Learning to "Everyone Else"," 2024.
- [15] S. Raisch and S. Krakowski, "Artificial Intelligence and Management: The Automation–Augmentation Paradox," *Academy of Management Review*, vol. 46, pp. 192–210, Jan. 2021.
- [16] M. H. Jarrahi, "Artificial intelligence and the future of work: Human-AI symbiosis in organizational decision making," *Business Horizons*, vol. 61, pp. 577–586, July 2018.
- [17] I. Seeber, E. Bittner, R. O. Briggs, T. De Vreede, G.-J. De Vreede, A. Elkins, R. Maier, A. B. Merz, S. Oeste-Reiß, N. Randrup, G. Schwabe, and M. Söllner, "Machines as teammates: A research agenda on AI in team collaboration," *Information & Management*, 2020.
- [18] J. Janssen, B. Châtel, N. Den Heijer, R. Tieben, M. Deen, R. Corten, G. Peeters, and M. Olde Rikkert, "A Digital Gaming Intervention to Strengthen the Social Networks of Older Dutch Adults," *JMIR Formative Research*, 2023.
- [19] C. J. Cai, S. Winter, D. Steiner, L. Wilcox, and M. Terry, "'Hello AI': Uncovering the Onboarding Needs of Medical Practitioners for Human-AI Collaborative Decision-Making," *Proceedings of the ACM on Human-Computer Interaction*, vol. 3, pp. 1–24, Nov. 2019.
- [20] B. Shneiderman, "Human-centered AI: ensuring human control while increasing automation," in *Proceedings of the 5th Workshop on Human Factors in Hypertext*, (Barcelona Spain), pp. 1–2, ACM, June 2022.
- [21] H. Dubberly and S. Evenson, "On modelingThe analysis-synthesis bridge model," *interactions*, vol. 15, no. 2, pp. 57–61, 2008.
- [22] B. Song, Q. Zhu, and J. Luo, "Human-AI collaboration by design," *Proceedings of the Design Society*, vol. 4, pp. 2247–2256, May 2024.
- [23] J. P. Guilford, "The structure of intellect," *Psychological Bulletin*, vol. 53, no. 4, pp. 267–293, 1956. Place: US Publisher: American Psychological Association.
- [24] K. Peffers, T. Tuunanen, M. Rothenberger, and S. Chatterjee, "A Design Science Research Methodology for Information Systems Research," *J. Manage. Inf. Syst.*, vol. 24, pp. 45–77, Dec. 2007.
- [25] R. T. Watson, , and J. Webster, "Analysing the past to prepare for the future: Writing a literature review a roadmap for release 2.0," *Journal of Decision Systems*, pp. 129–147, 2020.
- [26] L. A. Palinkas, S. M. Horwitz, C. A. Green, J. P. Wisdom, N. Duan, and K. Hoagwood, "Purposeful Sampling for Qualitative Data Collection and Analysis in Mixed Method Implementation Research," pp. 533–544, 2013.
- [27] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research," *MIS Q.*, vol. 28, pp. 75–105, Mar. 2004.
- [28] M. B. Miles, A. M. Huberman, and J. Saldaña, *Qualitative data analysis: a methods sourcebook*. Thousand Oaks, California: SAGE Publications, Inc, third edition ed., 2014.
- [29] E. Brynjolfsson, T. Mitchell, and D. Rock, "What Can Machines Learn, and What Does It Mean for Occupations and the Economy?," *AEA Papers and Proceedings*, vol. 108, pp. 43–47, May 2018.
- [30] T. H. Davenport, "From analytics to artificial intelligence," *Journal of Business Analytics*, vol. 1, pp. 73–80, July 2018.
- [31] P. Tambe, P. Cappelli, and V. Yakubovich, "Artificial Intelligence in Human Resources Management: Challenges and a Path Forward," *California Management Review*, pp. 15–42, 2019.
- [32] A. Maedche, C. Legner, A. Benlian, B. Berger, H. Gimpel, T. Hess, O. Hinz, S. Morana, and M. Söllner, "AI-Based Digital Assistants: Opportunities, Threats, and Research Perspectives," *Business & Information Systems Engineering*, vol. 61, pp. 535–544, Aug. 2019.
- [33] L. Bainbridge, "Ironies of automation," *Automatica*, vol. 19, pp. 775–779, Nov. 1983.
- [34] R. Parasuraman and D. H. Manzey, "Complacency and Bias in Human Use of Automation: An Attentional Integration," *Human Factors*, vol. 52, pp. 381–410, June 2010. Publisher: SAGE Publications Inc.
- [35] L. Wang, H.-C. Zhang, and Q. Wang, "On the Concepts of Artificial Intelligence and Innovative Design in Product Design," *IOP Conference Series: Materials Science and Engineering*, vol. 573, p. 012095, July 2019.
- [36] U. Wilkens, L. Immanuel, Z. Connie, B. Aplper, , and M. Prilla, "Augmenting diversity in hiring decisions with artificial intelligence tools," *The International Journal of Human Resource Management*, pp. 1–38, 2025. Publisher: Routledge _eprint: <https://doi.org/10.1080/09585192.2025.2492867>.
- [37] M. Vössing, N. Kühn, M. Lind, and G. Satzger, "Designing Transparency for Effective Human-AI Collaboration," *Information Systems Frontiers*, vol. 24, pp. 877–895, June 2022.
- [38] M. Pinski and A. Benlian, "AI Literacy - Towards Measuring Human Competency in Artificial Intelligence," 2023.
- [39] R. Annapureddy, A. Fornaroli, and D. Gatica-Perez, "Generative AI Literacy: Twelve Defining Competencies," *Digit. Gov.: Res. Pract.*, vol. 6, pp. 13:1–13:21, Feb. 2025.
- [40] D. Long and B. Magerko, "What is AI Literacy? Competencies and Design Considerations," *CHI '20*, pp. 1–16, Association for Computing Machinery, 2020.